Siddharth Roy

siddharthroy2708@gmail.com | LinkedIn | GitHub | +91 6200552175

Education

National Institute of Technology, Jamshedpur — August 2023 – Present
B.Tech in Electronics And Communication Engineering — *Jamshedpur, Jharkhand*

Indian High School, Dubai — April 2022 – May 2023
12th Grade, 75.6% — *Dubai, U.A.E.*

Birla Public School, Pilani — April 2020 – May 2021
10th Grade, 88.4% — *Pilani, Rajasthan*

Domains / Fundamentals

Languages: Python, C/C++, TypeScript, JavaScript, Bash, HTML/CSS, SQL
Frameworks & Libraries: Node.js, Next.js, React.js, TensorFlow, PyTorch, NumPy, Pandas
ML/AI: Machine Learning, Deep Learning, Computer Vision, Feature Engineering
Systems & OS: Linux, Shell Scripting, Operating Systems, Computer Networks
Developer Tools: Git/GitHub, VS Code, PyCharm, CMake, Docker, Google Cloud Platform
Relevant Coursework: OS, IoT, Data Structures and Algorithms, AI/ML, Soft Computing.

Experience

AI/ML Intern — May 2025 – July 2025
ROBOMANTHAN — *Link*

- Developed **computer vision and ML applications** on Raspberry Pi using OpenCV and Python for real-time object detection, image processing, and autonomous perception.
- Integrated ML models with ROS frameworks to enable **autonomous navigation, obstacle detection, and tracking** for robotics research platforms.

Projects

Multi-threaded Reverse Proxy Server — GitHub |
|*C++, epoll, Thread Pool, HTTP Parsing, Load Balancing, LRU Cache* |

- **Engineered** a high-performance reverse proxy handling **5,000+ concurrent connections** with epoll-based I/O, custom thread pool, and asynchronous logging.
- Implemented **HTTP parsing**, Round-Robin and Least-Connections load balancing, **LRU caching**, and backend failover in a modular, scalable architecture inspired by Nginx.

NoteForge-AI: AI-Enhanced Notion-Style Notes App — GitHub |
|*Next.js 15, TypeScript, PostgreSQL/Prisma, OpenAI API, WebSockets, Stripe* |

- **Built** a full-stack SaaS notes platform with AI assistance, enabling **100+ concurrent users** and **50% faster note creation** via AI summarization and text generation.
- **Integrated** authentication, Stripe subscriptions, PostgreSQL/Prisma DB, and WebSocket-based real-time sync for **scalable production deployment**.

Distributed Key-Value Store (Raft-Based) — GitHub |
|*Go, gRPC, Protobuf, Raft Consensus* |

- **Built** a fault-tolerant distributed KV store with **3–5 node clusters**, ensuring **100% data consistency** using Raft consensus and leader election.
- **Implemented** log replication, heartbeats, and gRPC/Protobuf RPCs with a WAL-based storage engine for **safe concurrent operations** and resilient cluster management.

Achievements

- **E-Summit (E-Cell):** Won the Chat JPG event, a competitive Prompt Engineering-based challenge.
- **Ojass 2025:** Achieved 2nd place in the 7kg Robo War, demonstrating combat robotics skills.

Positions of Responsibility

- **Vice President** – Society of Electronics and Communication Engineering, NIT Jamshedpur.
- **Head of Electronics and Avionics** – Cosmology And Rocketry Club, NIT Jamshedpur.